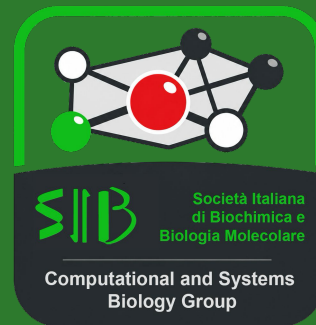


S1 · 10:25–10:40

Expanding the iMeGroCy Framework: Dynamic Nutrient Responses, Ribosome Biogenesis Modules, and Mammalian Cell Modeling

Pasquale Palumbo

University of Milano-Bicocca



Expanding the iMeGroCy Framework: Dynamic Nutrient Responses, Ribosome Biogenesis Modules, and Mammalian Cell Modeling

Marco Vanoni^{1,2}, Pasquale Palumbo^{1,2}, Federico Papa³, Stefano Busti^{1,2},
Lilia Alberghina^{1,2}

¹ *Department of Biotechnology and Biosciences, University of Milano-Bicocca, Milan, Italy*

² *SYSBIO, Centre of Systems Biology, Milan, Italy*

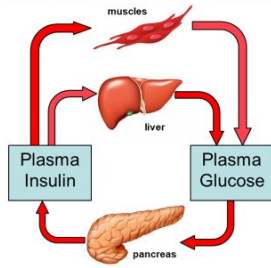
³ *Institute for Systems Analysis and Computer Science "A. Ruberti" (IASI), CNR, Rome, Italy*



pasquale.palumbo@unimib.it

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What about me



Glucose-insulin models and the Artificial Pancreas

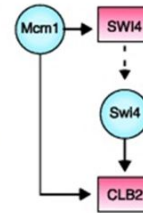


Diabetes progression



Tumor growth control

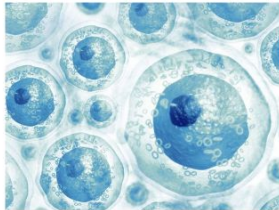
Feedforward Loop



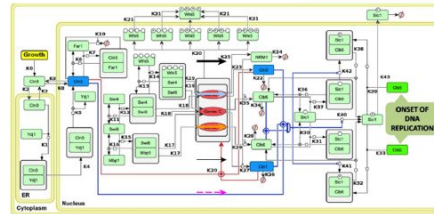
Transcription networks



Microbial communities



Stem cell proliferation



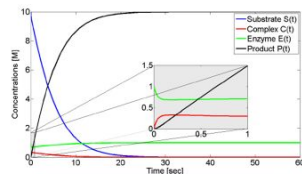
Chemical Reaction Networks

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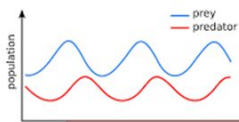
What about me

$$\begin{aligned}\frac{dn_1}{dt} &= (k_1 - k_2)n_1 - k_5 n_1 \frac{n_3}{M_5 + n_3}, \\ \frac{dn_2}{dt} &= k_2 n_1 - k_3 n_2 + k_5 n_1 \frac{n_3}{M_5 + n_3}, \\ \frac{dn_3}{dt} &= -\rho \frac{n_3}{M_4 + n_3} - k_5 n_1 \frac{n_3}{M_5 + n_3} + r,\end{aligned}$$

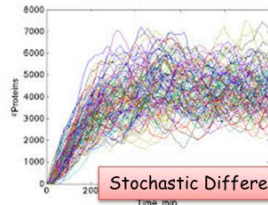
Ordinary Differential Equations (ODE)



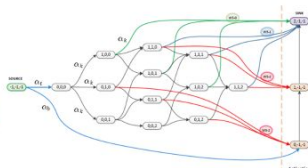
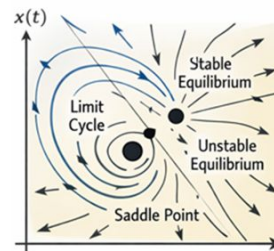
Double-time scale (QSSA)



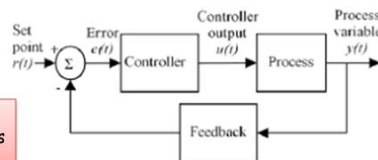
Lotka-Volterra models



Stochastic Differential Equations (SDE)

Chemical Master Equations
Continuous-Time Markov Chains
Monte Carlo simulations

Qualitative behaviour analysis



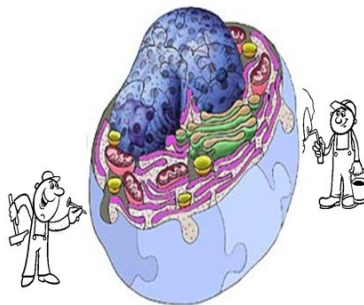
Feedback control

Towards *whole cell* models

Quantitative **whole-cell models**, striving to account for as many biological functions as possible, are required to explain **complex phenotypes** and to accelerate biological discoveries

CHALLENGES:

- different time-scales
- different layers (hierarchical modeling)
- different levels of graining to be explained
- different kinds of data (steady-state/time-varying measurements, interactomics, metabolomics, etc.)

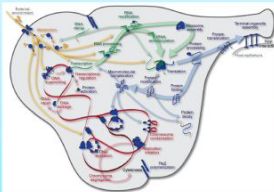


Building up whole-cell models is a complex procedure requiring to adopt **different mathematical tools** and a **large amount of data**

Towards yeast whole cell models

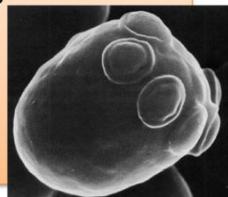
Breakthrough paper: Karr et al, Cell, 2012

- a first success in whole-cell modeling for *M. genitalium* (in terms of model capabilities to replicate available data)
- a very basic microorganism
 - ✓ bacterium
 - ✓ about 500 genes
 - ✓ no signaling pathways
 - ✓ no adaptation to external cues



S. cerevisiae is a much more complicated microorganism

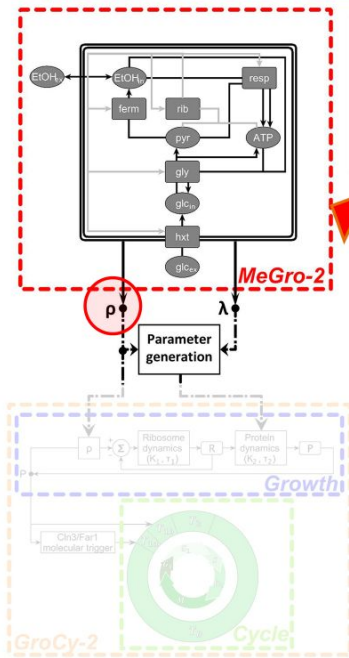
- eukaryotic cell with a compartmentalized cellular organization
- ten-fold larger genome
- different signaling pathways
- large set of homeostatic response to external cues
- sophisticated nutritionally modulated sensing



Multiscale integration: *iMeGroCy* in a nutshell

In Vanoni et al, 2025 we build up a **scaffold** providing a **coarse-grain** representation of the functions essential for the life of yeast:
i) **metabolism** ii) **growth** iii) **cell cycle**

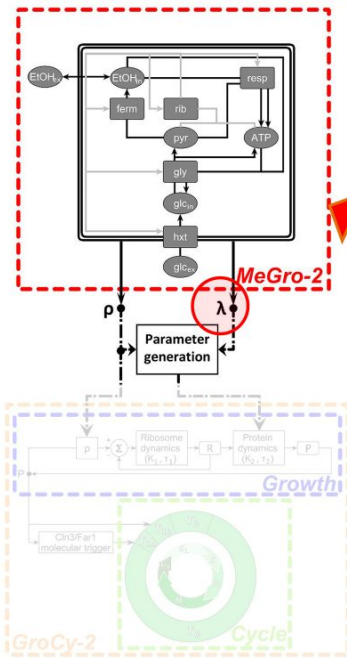
The *MeGro* module exploits the available nutrients and other data from experiments as inputs to compute the **exponential growth rate λ**



MeGro models the metabolism fast dynamics, treated as an algebraic constraint

Multiscale integration: *iMeGroCy* in a nutshell

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The *MeGro* module exploits the available nutrients and other data from experiments as inputs to compute the **exponential growth rate λ**

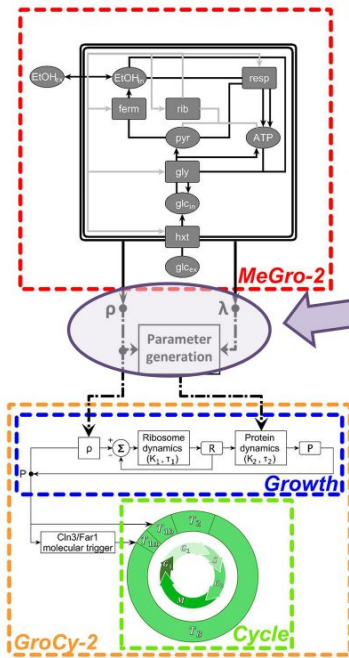
In addition, *MeGro* also provides the **ribosome-over-protein ratio p**

MeGro models the metabolism fast dynamics, treated as an algebraic constraint

Multiscale integration: *iMeGroCy* in a nutshell

In Vanoni et al, 2025 we build up a **scaffold** providing a **coarse-grain** representation of the functions essential for the life of yeast:
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λ and ρ allow to compute the protein and ribosome dynamics parameters as inputs for the *GroCy* module

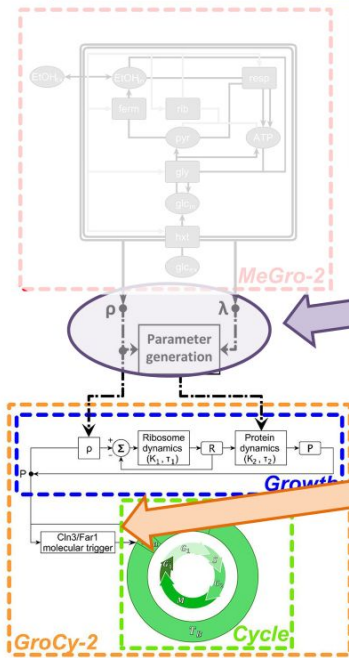


Multiscale integration: *iMeGroCy* in a nutshell

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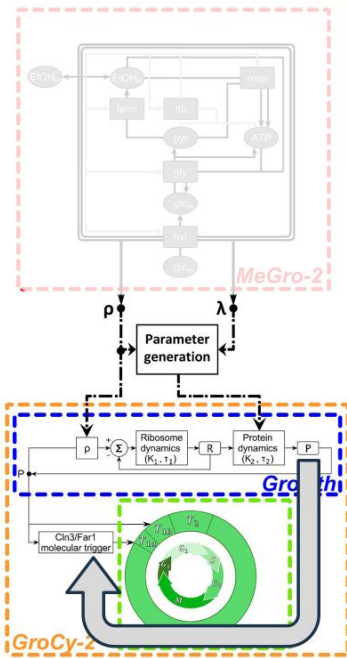
λ and ρ allow to compute the protein and ribosome dynamics parameters as inputs for the *GroCy* module

Besides protein and ribosome dynamics, the Growth module involves a minimal set of molecular players, modeling the whole as a set of ODE



Multiscale integration: *iMeGroCy* in a nutshell

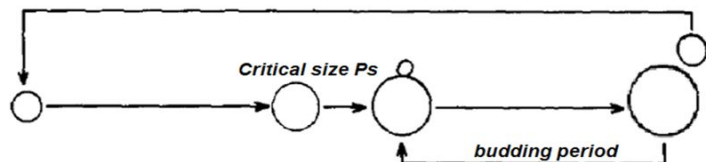
In [Vanoni et al, 2025](#) we build up a **scaffold** providing a **coarse-grain** representation of the functions essential for the life of yeast:
i) **metabolism** ii) **growth** iii) **cell cycle**



The accumulation of specific molecular players described by the Growth ODE module triggers the initiation of cell cycle events as a set of consecutive Timers

GroCy: cellular division

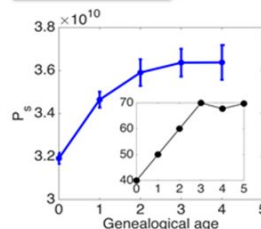
- The molecular players (in the nucleus) are equally divided between newborn daughter and parent cells
- The initial protein content of the old parent cell is given by P_s (the critical size); the initial protein content of the newborn daughter cell is given by the proteins synthesized during the budded period
- Ribosomes are divided proportionally to the protein content



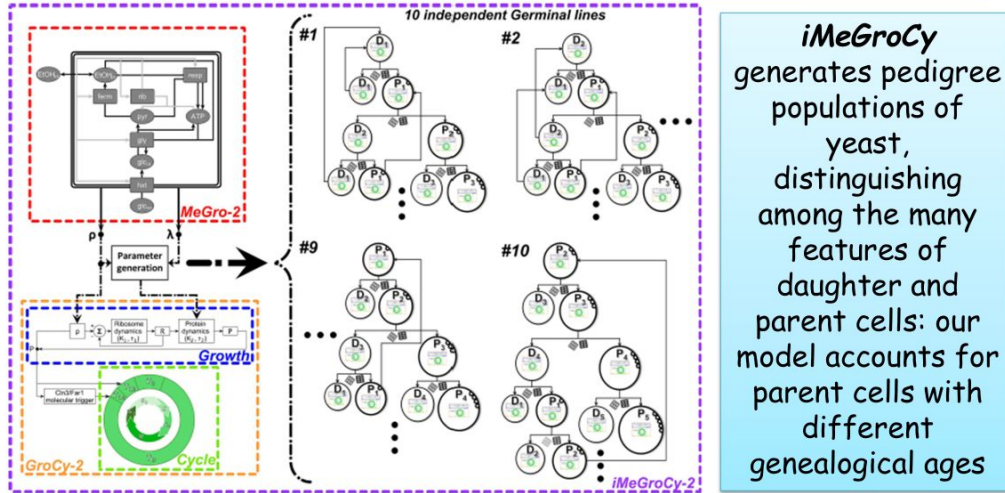
newborn D

newborn P

After each division, parent cells increase in size before the bud: hence, parent cell mass at budding increases with genealogical age



iMeGroCy: from single cells to populations



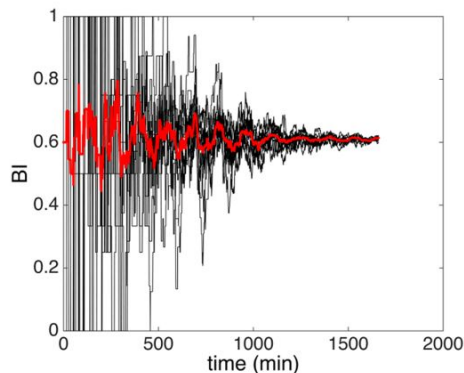
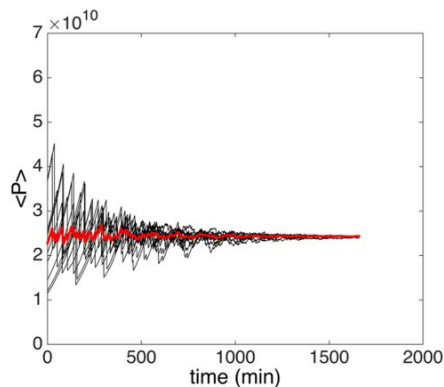
iMeGroCy generates pedigree populations of yeast, distinguishing among the many features of daughter and parent cells: our model accounts for parent cells with different genealogical ages

Inter-cellular variability is considered only for a reduced set of model parameters (Timer lengths and parent/daughter protein/ribosome partition at division)

Populations of *iMeGroCy*: outputs

Population variables that evolve with time:

- Total (average) protein/ribosome content
- Number of cells
- Budded index (fraction of budded cells)



Populations of *iMeGroCy*: outputs

Statistics achieved on a pool of cells:

- Timer lengths
- Size at birth, at START, at division

These outputs can be computed on specific types of cells (i.e. Daughters/Parents, Parents of given age, budded/unbudded, etc.)

Protein distributions

- Achieved for cells selected at a given time instant
- Provided for Parent/Daughter or for the whole populations
- Closely related to the age distribution

Populations of *iMeGroCy*: outputs

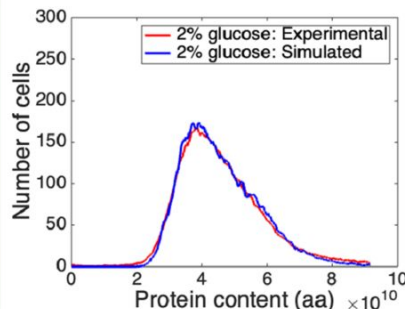
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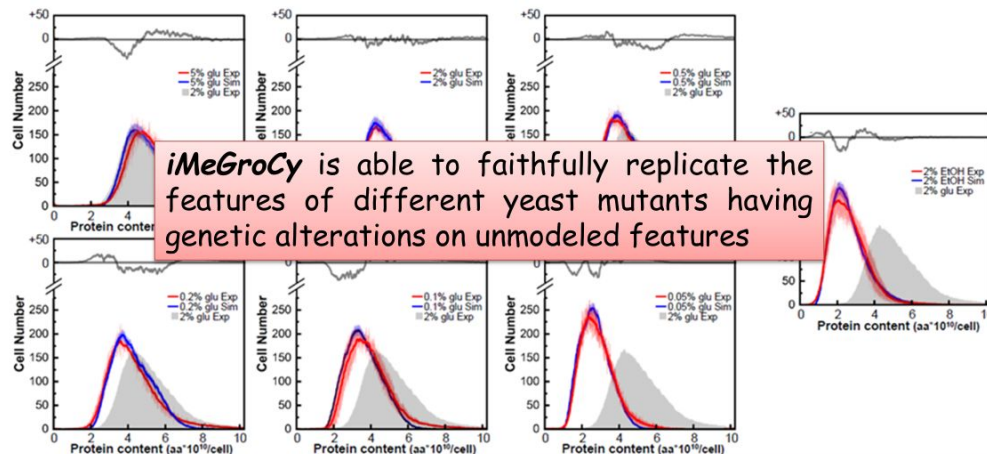
Protein distributions

- Achieved for cells selected at a given time instant
- Provided for Parent/Daughter or for the whole populations
- Closely related to the age distribution
- **Protein distributions indicate how a population is growing at the time of analysis**



Results (1)

An almost unique set of model parameters is able to fit different nutritional environments: MeGro processes the nutritional information to feed the GroCy population automatically

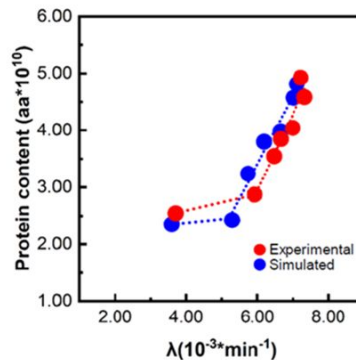


pasquale.palumbo@unimib.it

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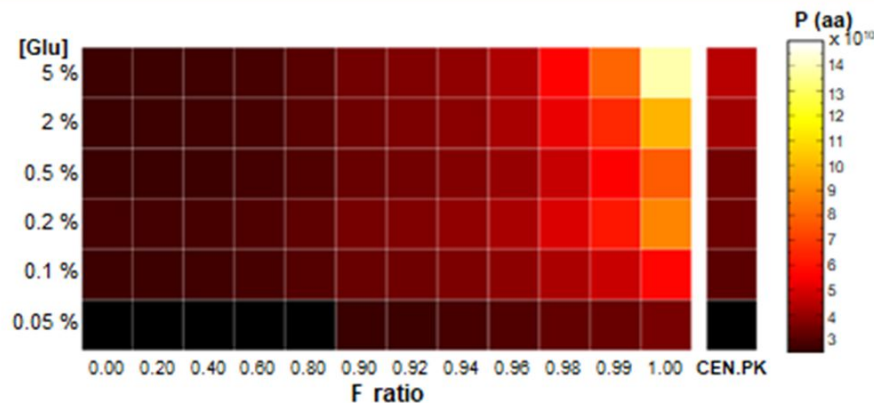
Results (2)

Metabolism acts as the master controller of the growth and cycle modules: for both experimental and computational distributions, **glucose concentration strongly modulates the average protein content**, which shows a steep linear correlation with the growth rate λ



Results (3)

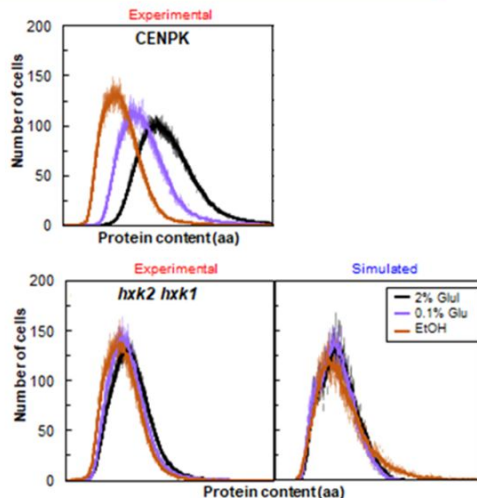
Fermentation is required to modulate cell size: The nutritionally-dependent modulation of the protein content occurs only when glucose metabolism is strongly shifted toward fermentation, being abolished when cells are "forced" to adopt a respiratory metabolism



Results (3): Validation using mutants

To validate these predictions, we analyzed the behavior of mutant *hxx2 hxx1*, defective in glucose metabolism (it adopts primarily respiratory metabolism)

As predicted by our model, in all the tested growth conditions, mutant cells adopt a small size regardless of the available carbon source, resulting in nearly superimposable protein distribution profiles



The diagram illustrates the integration of metabolic, signaling, and cell cycle models. The top left shows a metabolic map for MeGro-2, involving EtOH, ATP, and various metabolites. The top right shows a detailed signaling pathway from the cytoplasm to the nucleus, leading to the onset of DNA replication. The bottom left shows a cell cycle model with growth and cycle phases. The bottom right highlights the G1/S transition module.

MeGro-2

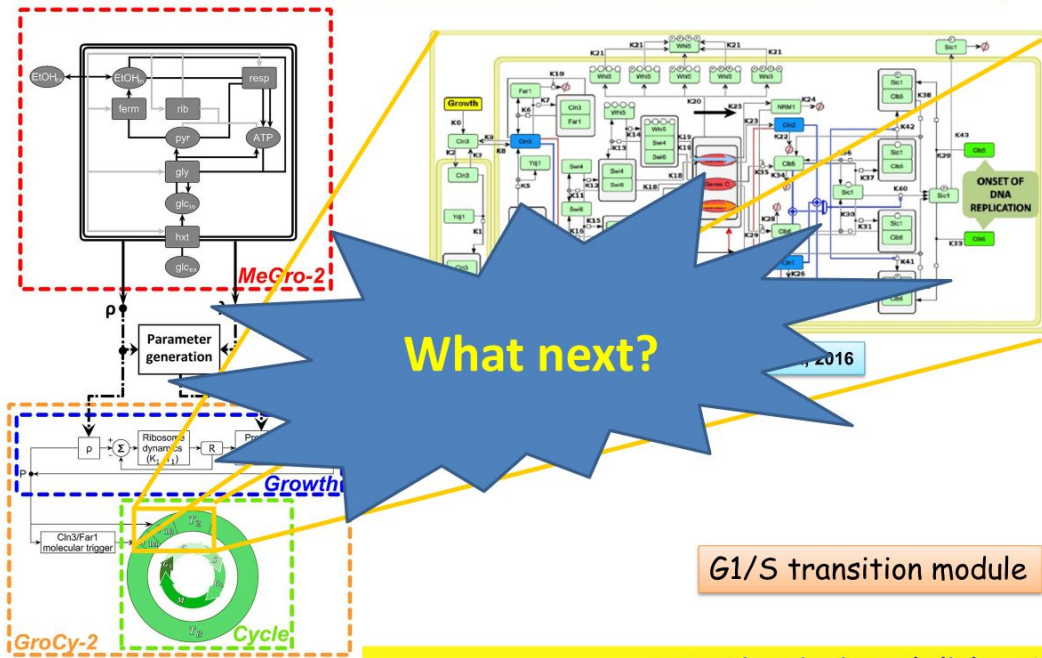
Palumbo et al, Nat Comm, 2016

GroCy-2

G1/S transition module

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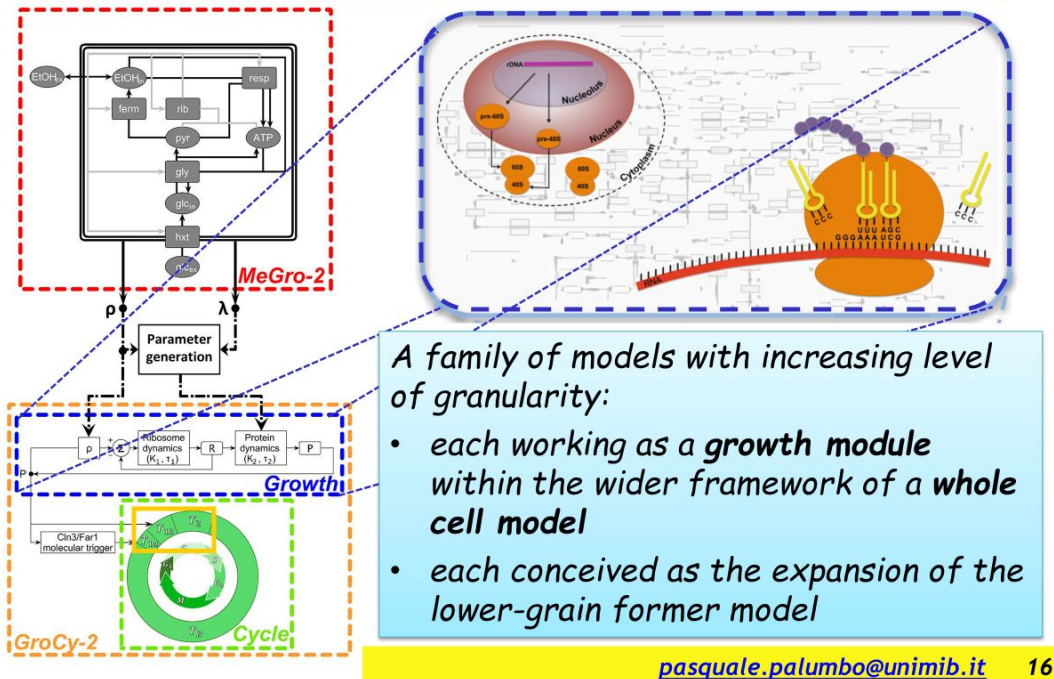
iMeGroCy as a scaffold for molecular plug-ins



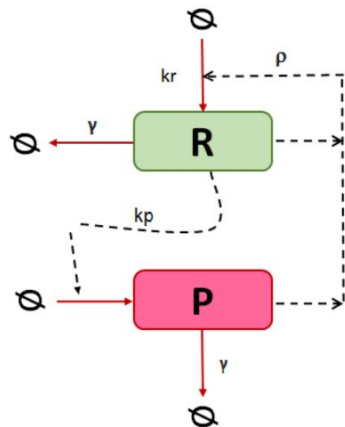
pasquale.palumbo@unimib.it

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iMeGroCy as a scaffold for Ribosome Biogenesis



Alberghina et al, Biosystems, 1986



$$\begin{aligned} \frac{dR_{\text{tot}}}{dt} &= k_r(\rho P - R_{\text{tot}})^+ - \gamma_{R_{\text{tot}}} R_{\text{tot}} \\ \frac{dP}{dt} &= k_p R_{\text{tot}} - \gamma_p P \end{aligned}$$

In this model, ribosome synthesis is switched off whenever the ribosome-over-protein ratio exceeds a given threshold ρ

Conditions on model parameters providing exponential grow

$$\lambda = \rho k_p - \gamma_p > 0$$

Palumbo et al, IFAC, 2023



Palumbo et al, IFAC, 2023

$$\frac{d\text{pre-35S}}{dt} = k_{\text{pre}} (\rho P - R_{\text{tot}})^+ - \beta_{\text{pre}} \cdot \text{pre-35S}$$

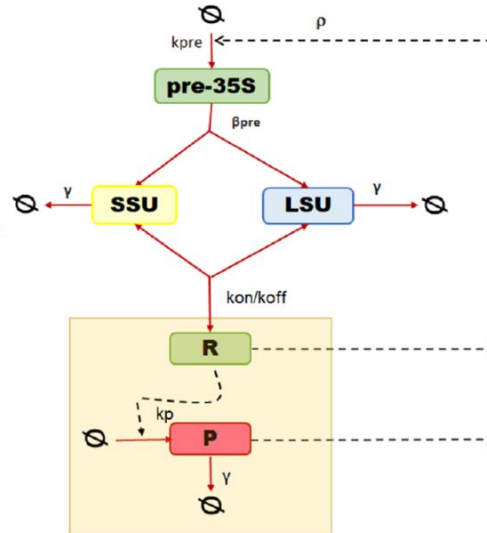
$$\frac{d\text{SSU}}{dt} = \beta_{\text{pre}} \cdot \text{pre-35S} - k_{\text{on}} \frac{\text{SSU} \cdot \text{LSU}}{P} + k_{\text{off}} \cdot R - \gamma_{\text{SSU}} \cdot \text{SSU}$$

$$\frac{d\text{LSU}}{dt} = \beta_{\text{pre}} \cdot \text{pre-35S} - k_{\text{on}} \cdot \frac{\text{SSU} \cdot \text{LSU}}{P} + k_{\text{off}} \cdot R - \gamma_{\text{LSU}} \cdot \text{LSU}$$

$$\frac{dR}{dt} = k_{\text{on}} \cdot \frac{\text{SSU} \cdot \text{LSU}}{P} - k_{\text{off}} \cdot R$$

$$\frac{dP}{dt} = k_p \cdot R - \gamma_p \cdot P$$

$$R_{\text{tot}} = R + \min\{\text{SSU}, \text{LSU}\}$$



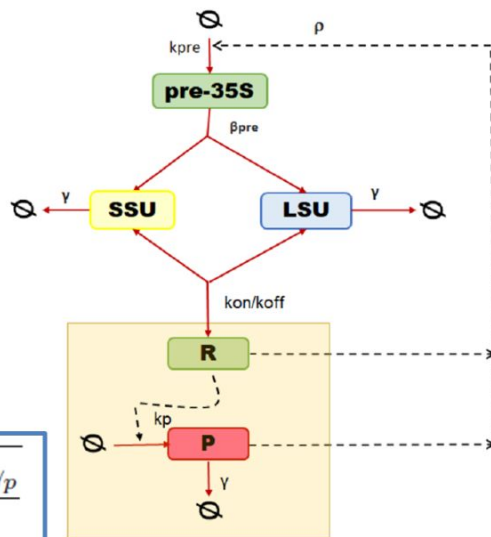
Palumbo et al, IEEE-CDC, 2024

Hp: Under the simplifying assumption that **SSU** and **LSU** are equimolar

Then:

- necessary + sufficient condition for exponential growth (i.e. all molecular players share the same exponential time law)

$$\rho k_p - \gamma_p > \left(1 + \frac{\gamma_{\text{SSU}}}{k_{\text{pre}}}\right) \sqrt{\frac{k_{\text{off}} k_p \gamma_p}{k_{\text{on}}}}$$



Palumbo et al, IEEE-CDC, 2024

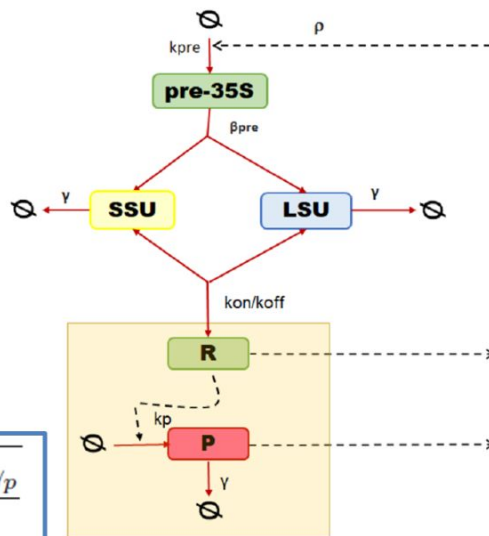
Hp: Under the simplifying assumption that **SSU** and **LSU** are equimolar

Then:

- necessary + sufficient condition for potential growth (all players share the same exponential time law)

$$\rho k_p - \gamma_p > 0$$

$$\rho k_p - \gamma_p > \left(1 + \frac{\gamma_{\text{SU}}}{k_{\text{pre}}}\right) \sqrt{\frac{k_{\text{off}} k_p \gamma_p}{k_{\text{on}}}}$$



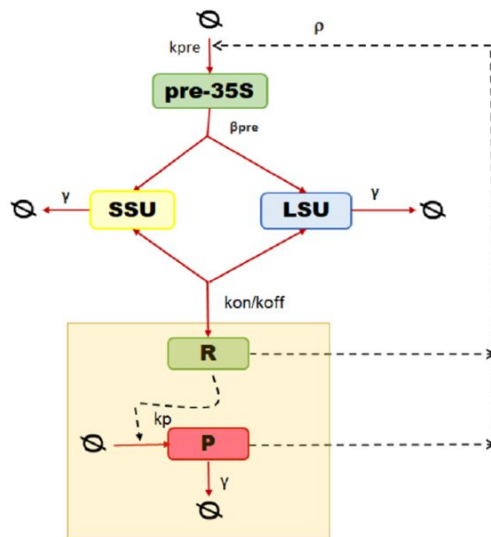
Palumbo et al, IEEE-CDC, 2024

Hp: Biologically meaningful simplifying assumption that *SSU* and *LSU* are equimolar

Then:

- necessary + sufficient condition for exponential growth (i.e. all molecular players share the same exponential time law)

$$\lambda = k_p \mu^* - \gamma_p$$



Palumbo et al, IEEE-CDC, 2024

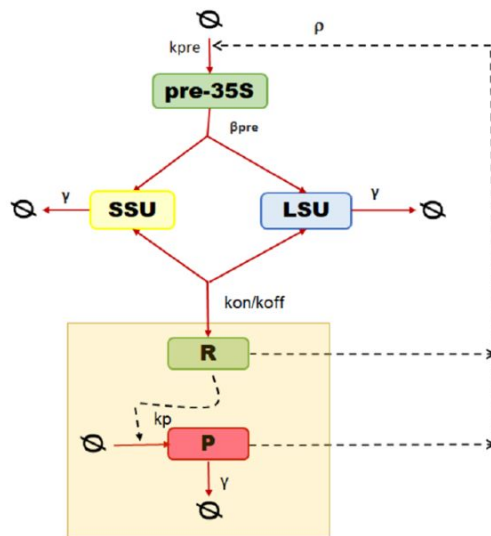
Hp: Biologically meaningful simplifying assumption that *SSU* and *LSU* are equimolar

Then:

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$$\lambda = k_p \mu^* - \gamma_p$$

$$\lambda = \rho k_p - \gamma_p$$

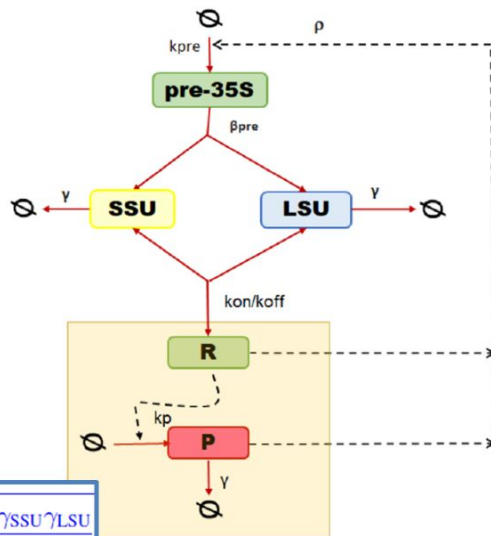


Palumbo et al, under review 2026

Hp: no simplifying assumption

Then:

- necessary + sufficient condition for exponential growth (i.e. all molecular players share the same exponential time law)



$$\rho k_p - \gamma_p > \left(\frac{1}{k_{pre}} + \frac{1}{\max\{\gamma_{SSU}, \gamma_{LSU}\}} \right) \sqrt{\frac{k_{off} k_p \gamma_p \gamma_{SSU} \gamma_{LSU}}{k_{on}}}$$

Palumbo et al, under review 2026

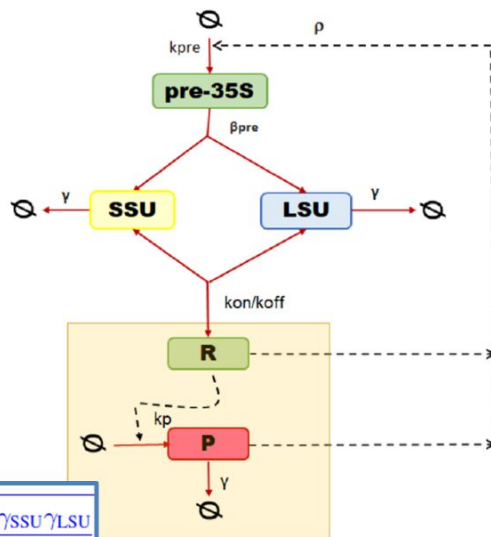
Hp: no simplifying assumption

Then:

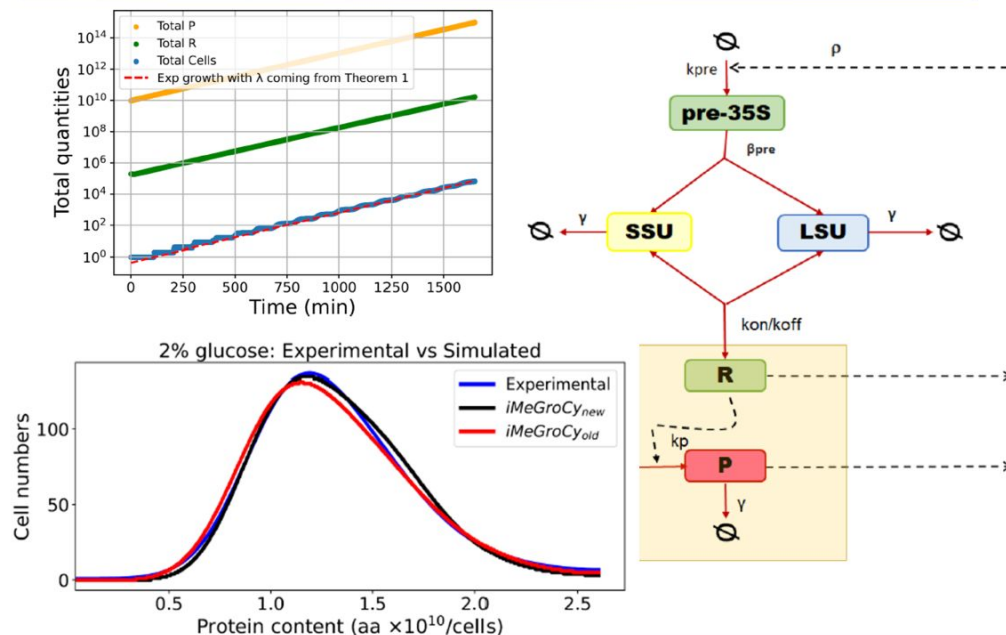
- necessary + sufficient condition for exponential growth (i.e. all molecular players share the same exponential time law)

$$\rho k_p - \gamma_p > \left(1 + \frac{\gamma_{\text{SU}}}{k_{\text{pre}}}\right) \sqrt{\frac{k_{\text{off}} k_p \gamma_p}{k_{\text{on}}}}$$

$$\rho k_p - \gamma_p > \left(\frac{1}{k_{\text{pre}}} + \frac{1}{\max\{\gamma_{\text{SSU}}, \gamma_{\text{LSU}}\}}\right) \sqrt{\frac{k_{\text{off}} k_p \gamma_p \gamma_{\text{SSU}} \gamma_{\text{LSU}}}{k_{\text{on}}}}$$

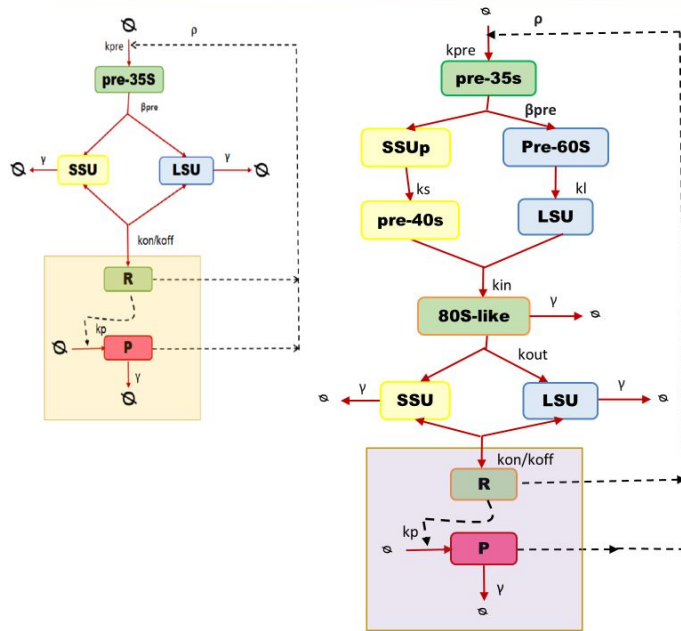


Palumbo et al, under review 2026


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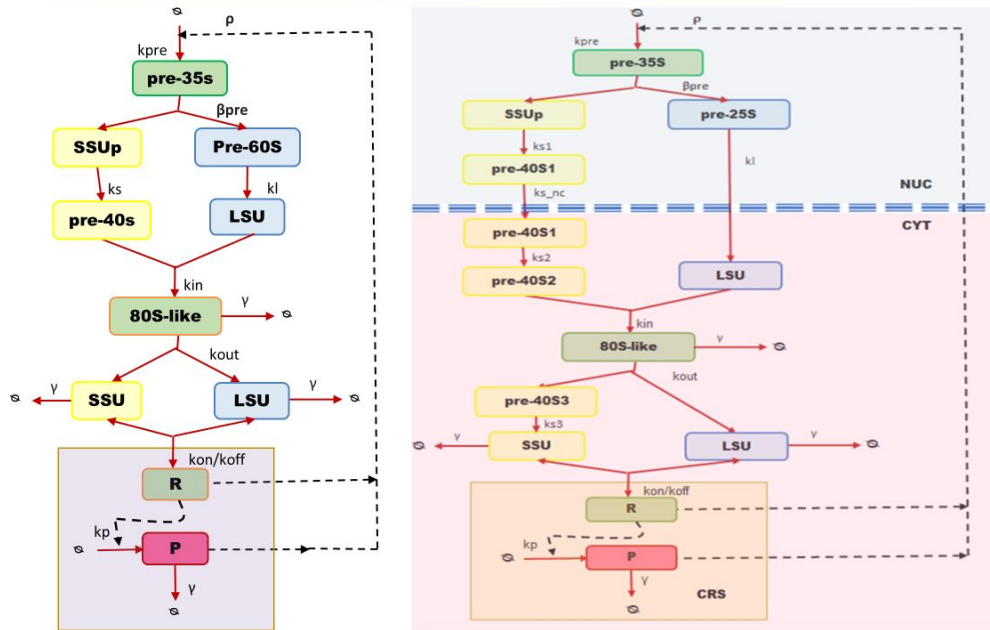
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Further developments...

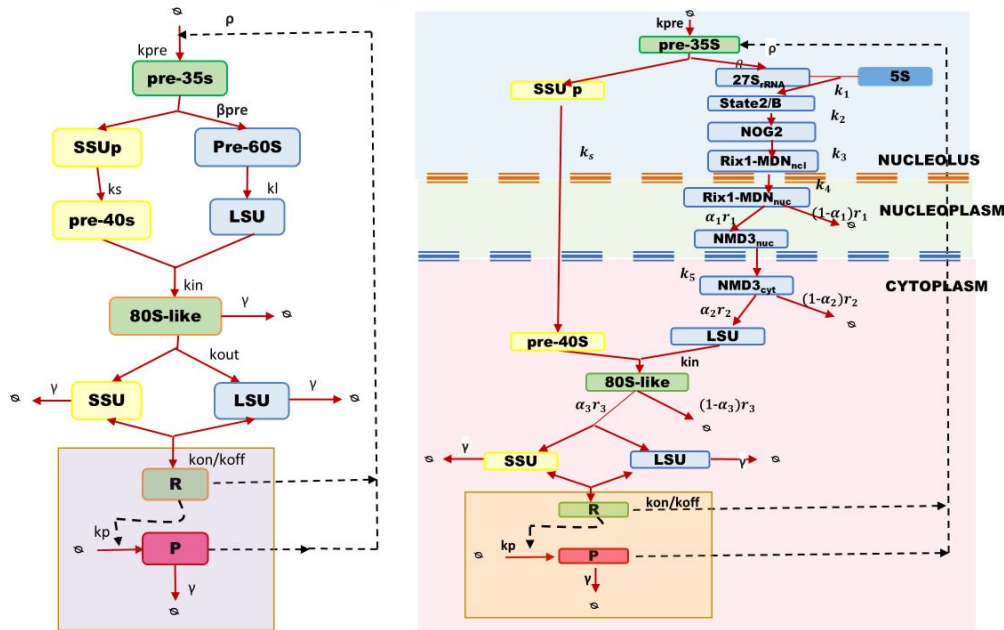


We extend the model by accounting for the SSU proofreading performed by the LSU

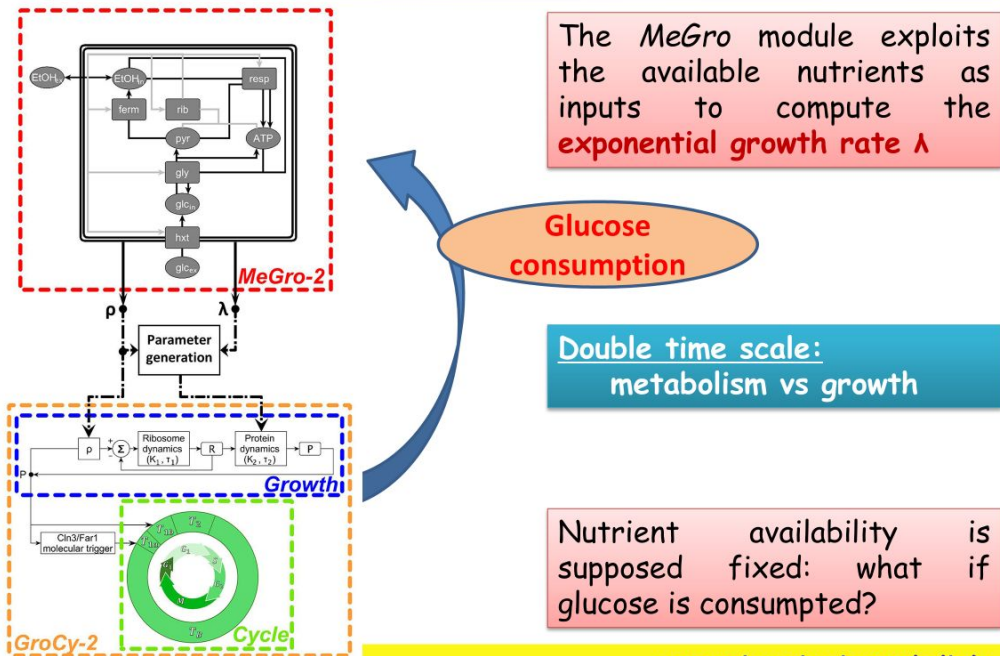
SSU is expanded in increasing details



LSU is expanded in increasing details



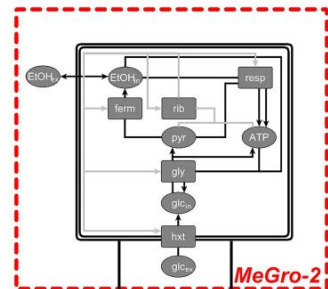
Nutrient responses: dynamic MeGro



pasquale.palumbo@unimib.it

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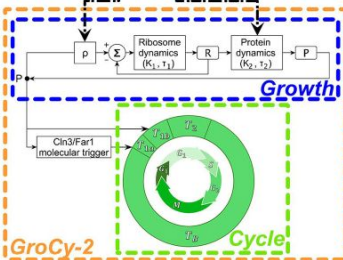
Nutrient responses: shift-up



A shift-up is the sudden transfer of cells from a nutrient-poor to a nutrient-rich environment (e.g. from ethanol to glucose)

Biological consequences:

- increased nutrient availability
- activation of growth-promoting pathways
- enhanced ribosome biogenesis
- increased protein synthesis
- faster cell-cycle progression

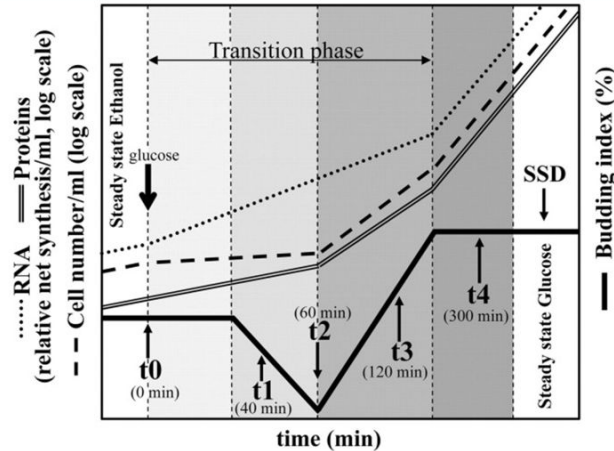


How do cells coordinate metabolic reprogramming with cell-cycle progression during the transition?

Nutrient responses: shift-up

The transient is modeled by rules:

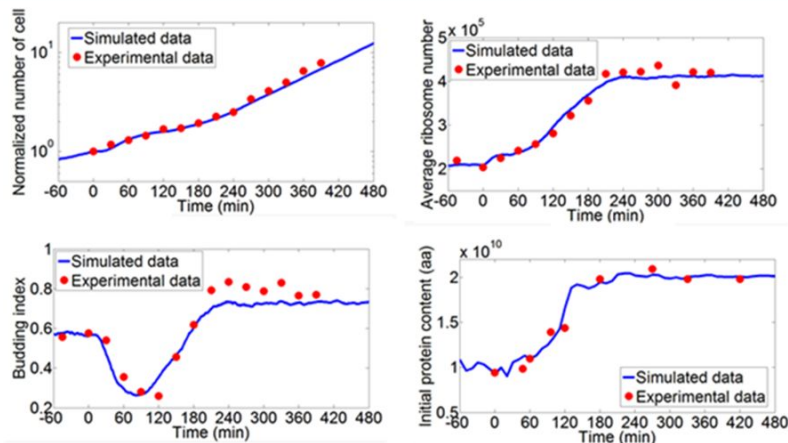
- Growth parameters change smoothly from slow to fast growth
- Cells in budded phase delay their mitosis
- Cells in G1 delay the G1/S transition



Nutrient responses: shift-up

The transient is modeled by rules:

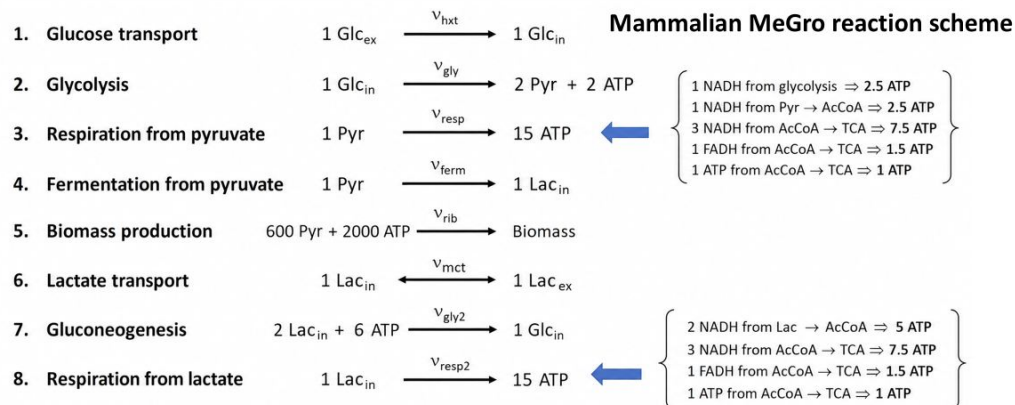
- Growth parameters change smoothly from slow to fast growth
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MeGro for mammalian cells

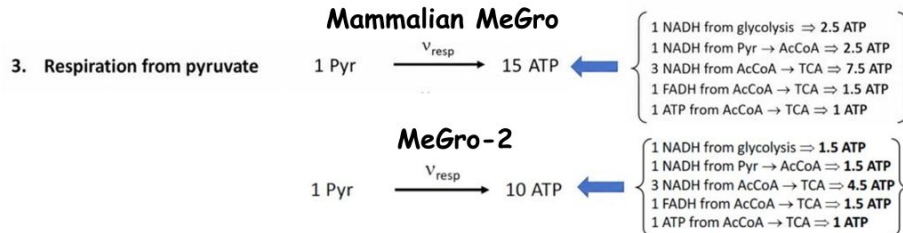
Understanding the metabolism-growth relationship in mammalian cells is fundamental for predicting cellular behavior in health and disease

The core metabolic structure is conserved across yeast and mammalian cells, yet specific reactions alter ATP yield



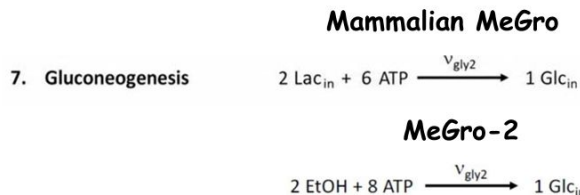
MeGro for mammalian cells

Mammalian cells possess complex I, resulting in greater proton translocation and higher ATP yield per NADH



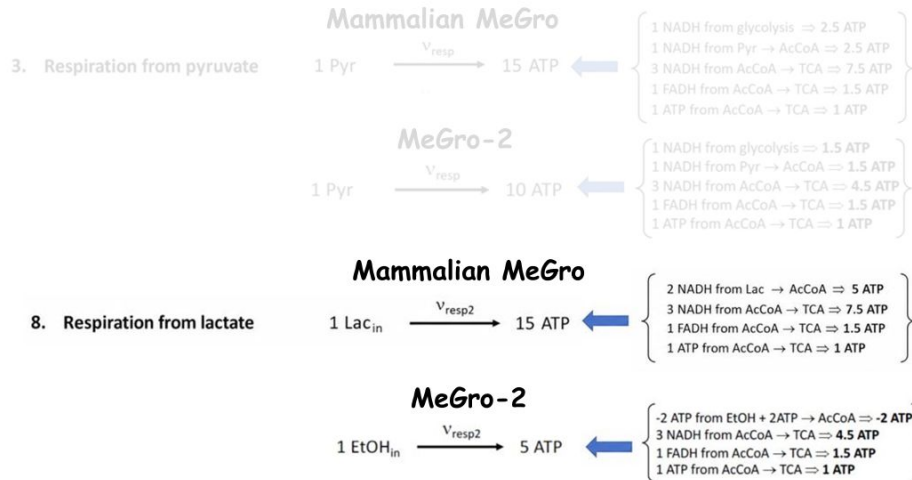
MeGro for mammalian cells

Yeast expends additional ATP to convert ethanol into acetyl-CoA, whereas mammalian cells convert lactate directly to pyruvate



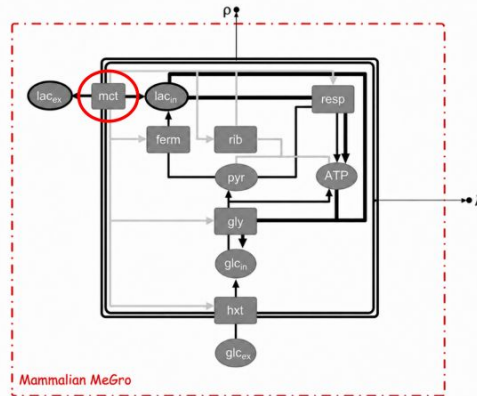
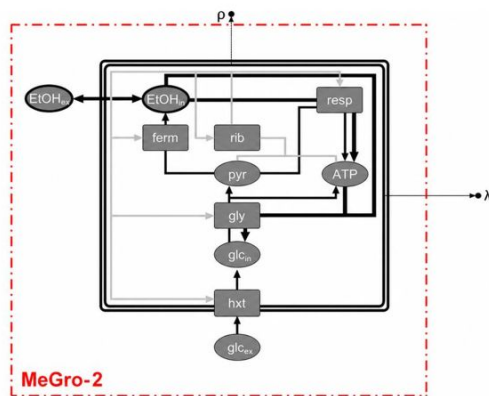
MeGro for mammalian cells

Yeast expends additional ATP to convert ethanol into acetyl-CoA, whereas mammalian cells convert lactate directly to pyruvate



MeGro for mammalian cells

Unlike ethanol diffusion, lactate exchange requires monocarboxylate transporters (MCTs), making transport flux dependent on transporter abundance



Conclusions

iMeGroCy provides:

- ✓ integration of metabolism, growth and cell cycle in budding yeast *Saccharomyces cerevisiae*
- ✓ prediction of population-level phenotypes
- ✓ modular scaffold for molecular plug-ins (G1/S transition, ribosome biogenesis, etc.)
- ✓ framework for dynamic nutrient responses (e.g. nutritional shifts)
- ✓ route toward mammalian whole-cell models

References



An Integrated Model Quantitatively Describing Metabolism, Growth and Cell Cycle in Budding Yeast

Pasquale Palumbo^{1,2(Corresponding)}, Marco Vanoni^{1,3}, Federico Papa^{1,2},
Stefano Busti^{1,3}, Meike Wortel^{1,5}, Bas Teusink⁴,
and Lilia Alberghina^{1,3}

¹ SYSBIO Centre for Systems Biology, Milan, Italy
pasquale.palumbo@ias1.cnr.it,
{marco.vanoni, lilia.alberghina}@unimib.it

² Institute for System Analysis and Computer Science
"Antonio Ruberti" – CNR, Via dei Taurini 19, Rome, Italy

³ Department of Biotechnology and Biosciences, University of Milano-Bicocca,
Piazza Della Scienza 2, Milan, Italy

⁴ Systems Bioinformatics, VU University,

De Boelelaan 1087, 1081 HV Amsterdam, The Netherlands

⁵ Centre for Ecological and Evolutionary Synthesis (CEES), The Department
of Biosciences, University of Oslo, Blindernveien 31, 0371 Oslo, Norway

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pasquale.palumbo@unimib.it

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RESEARCH ARTICLE

A modular model integrating metabolism, growth, and cell cycle predicts that fermentation is required to modulate cell size in yeast populations

Marco Vanoni^{1,2*}, Pasquale Palumbo^{1,2*}, Federico Papa^{1,3}, Stefano Busti^{1,2}, Laura Gotti^{1,2}, Melke Wortel⁴, Bas Teusink⁵, Ivan Orlandi^{1,2}, Alex Pessina^{1,2}, Cristina Alroldi^{1,2}, Luca Brambilla^{1,2}, Marina Val^{1,2}, Lilla Alberghina^{1,2*}

1 SYSBIO Centre for Systems Biology, Milan, Italy, **2** Department of Biotechnology and Biosciences, University of Milano-Bicocca, Milan, Italy, **3** IASI-CNR, Rome, Italy, **4** University of Amsterdam, Swammerdam Institute for Life Sciences, Amsterdam, The Netherlands, **5** Vrije Universiteit Amsterdam, Systems Biology Lab, AIMMS, Amsterdam, The Netherlands

* These authors contributed equally to this work.

* marco.vanoni@unimib.it (MV); lilla.alberghina@unimib.it (LA)



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pasquale.palumbo@unimib.it

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pasquale.palumbo@unimib.it

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